TITLE:

Brute force attack detection using Microsoft sentinel

Objective:

To simulate and detect unauthorized login attempts using Microsoft sentinel.

Tools used:

>Microsoft Sentinel

>Azure Virtual Machine

>Log Analytics Workspace

>KQL

Scenario

A brute-force attack was simulated by entering incorrect passwords multiple times on a windows VM.

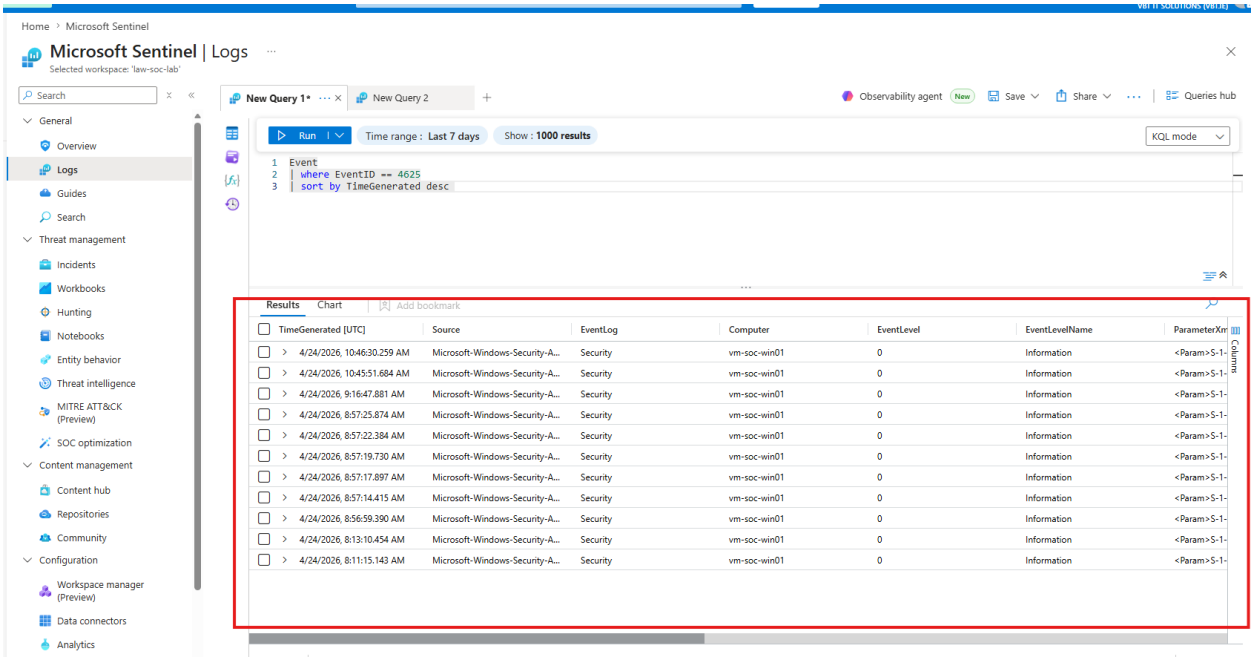
Detection:

The following query was used to detect failed login attempts:

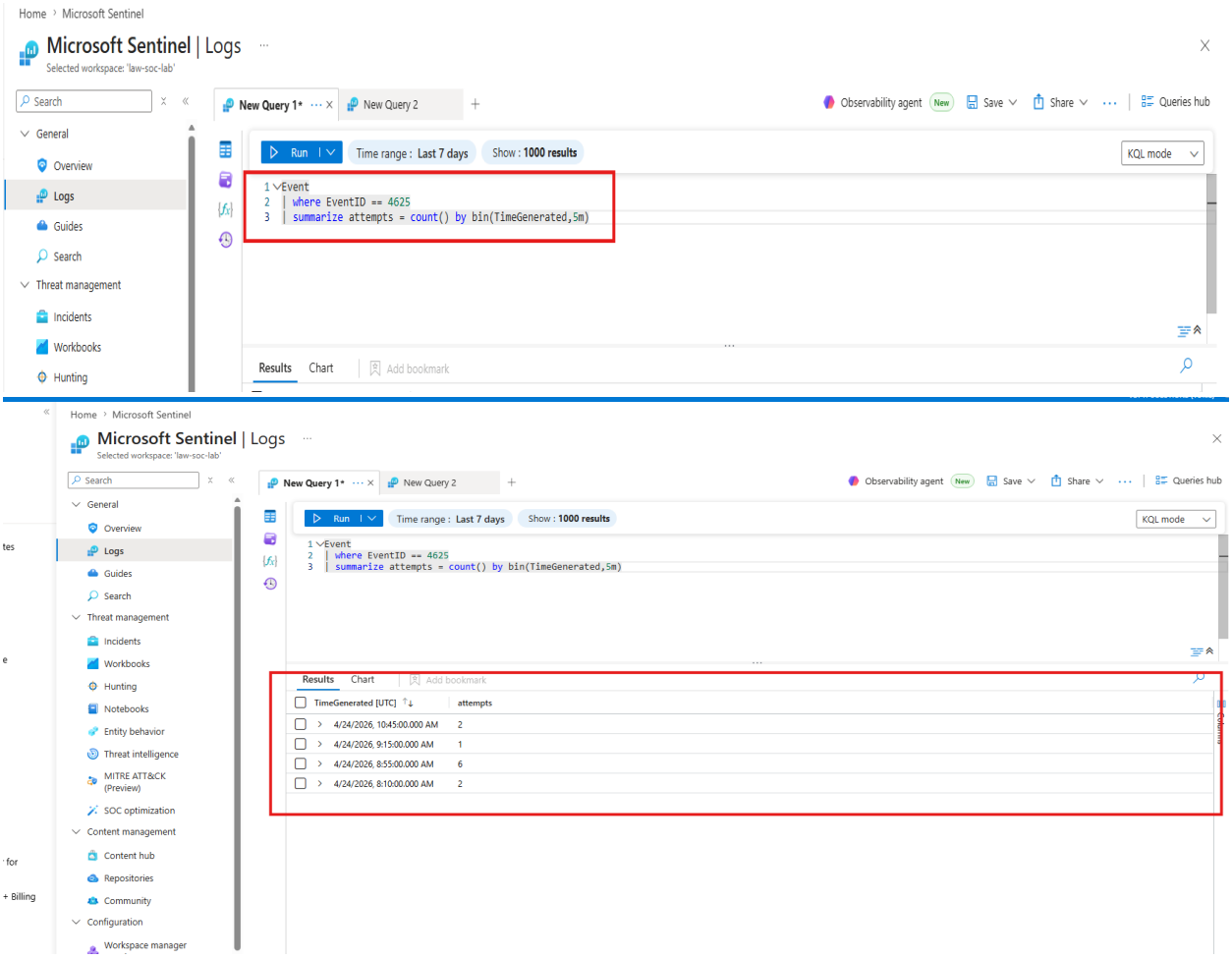
```
Event

| where EventID == 4625
```

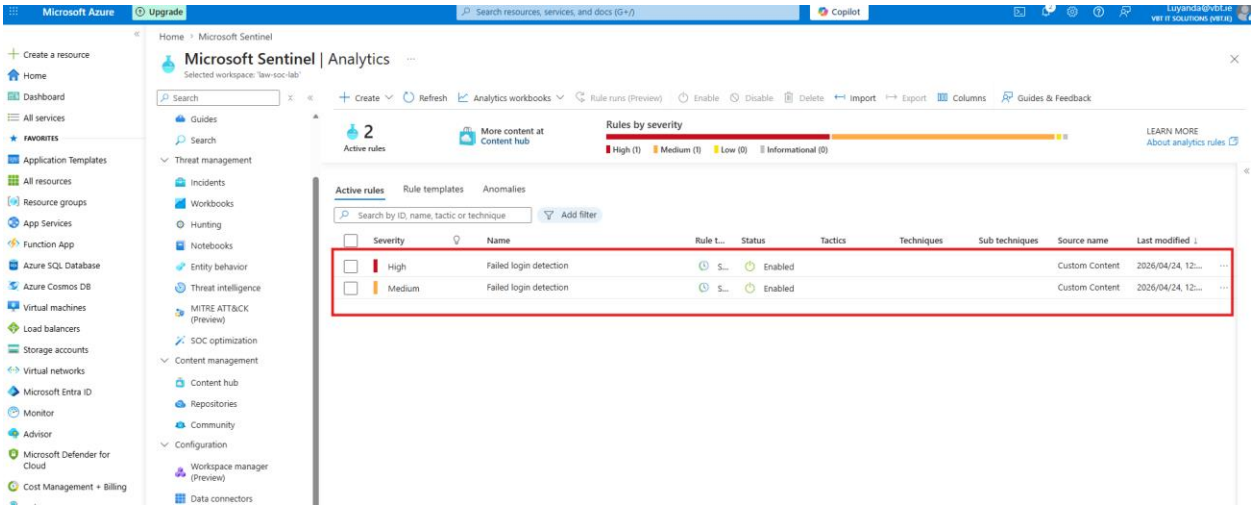
Detection screenshot

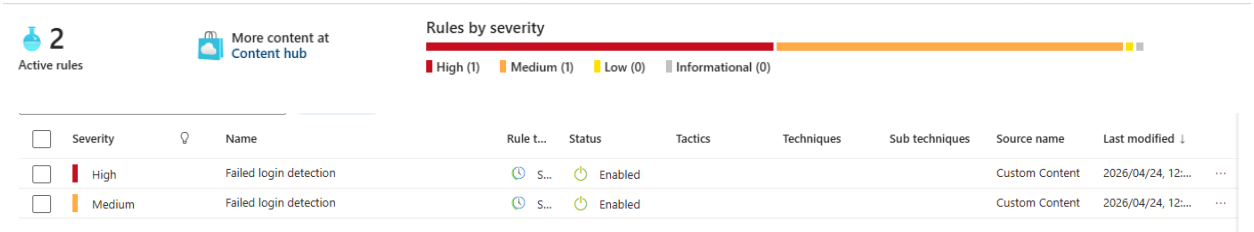


Analysis screenshots



Detection rule screenshot





Investigating findings:

>A total of 11 **failed login attempts** was detected within a short time window targeting the VM **vm-soc-win01**

>The failed attempts originated from 192.0.2.45, indicating a potential brute-force attack

>The rapid succession of login failures suggests automated attack behavior rather than normal user error.

>No successful login was observed following the failed attempts, reducing the likelihood of immediate compromise, but indicating an active attack attempt

Severity assessment:

This activity is classified as Medium to high severity, as repeted failed login attempts indicate a potential brute-force attack that could lead to unauthorized access if successful.

MITRE ATT&CK Mapping

>Technique: Brute force

>ID: T1110

Attack Timeline

>Initial failed login detected at 08:55

>Multiple failed attempts observed within 5 minutes

>No successful authentication recorded

Response

>Block suspicious IP address

>Enforce account lockout policy

>Enable Multi-Factor Authentication

>Monitor for further suspicious login attempts

>Review logs for a potential successful compromise

Recommendations

>Enable account lockout policies

>Implement multi-Fator authentication

>Monitor login anomalies continuously

Key takeaway

Successfully built a SOC lab, simulated attack, and implemented automated detection using Microsoft Sentinel.